DESIGN TOPOLOGIES

DESIGN & LAYOUT OF CMOS BANDGAP VOLTAGE REFERENCE USING SKY 130 PDK

Submitted by
Group 4
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Basic Concepts

PTAT - Proportional to Absolute Temperature (Increases as temperature increases)

$V_{PTAT} = V_T \ln(n)$

Term	Meaning	Explanation
V_{PTAT}	PTAT voltage	Voltage that increases with temperature. PTAT means Proportional To Absolute Temperature.
V_T	Thermal voltage	Thermal voltage (V_T) is the voltage associated with the thermal energy of charge carriers inside a semiconductor. $V_T = kT/q$ Proportional to Absolute temperature *Important: V_T is not the threshold voltage of a MOSFET. Thermal voltage and MOSFET threshold voltage are two different quantities. *
n	Emitter-area or current-density ratio	The ratio between two BJT emitter areas or, under suitable conditions, their current densities.
ln(n)	Natural logarithm of n	A mathematical factor that determines the magnitude of the PTAT voltage.
k	Boltzmann constant	$1.38 \times 10^{-23} J/K$
T	Absolute temperature	Temperature measured in kelvin (K).
q	Electron charge	$1.602 \times 10^{-19} C$

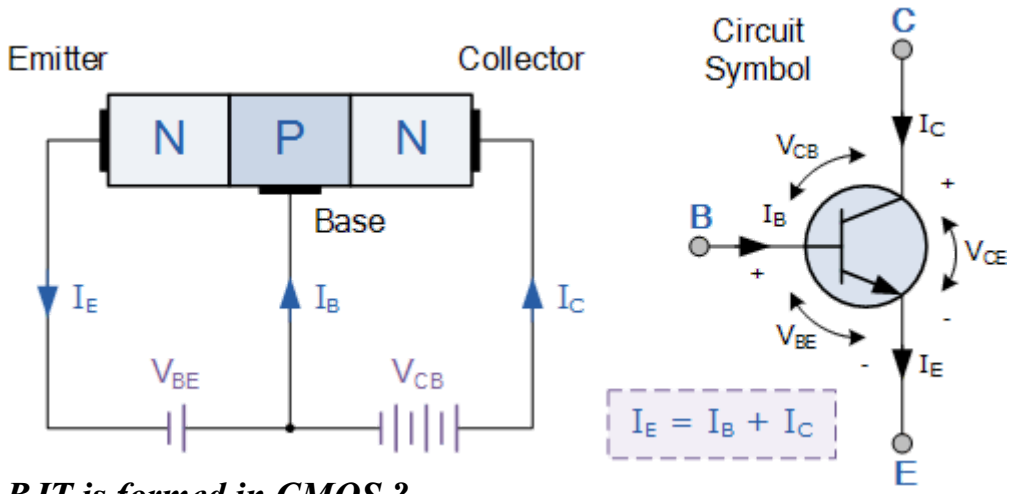
CTAT - Complementary to Absolute Temperature (Decreases as temperature increases)

$|V_{BE}|$ *generally decreases with temperature* = generated from a transistor with less emitter area (have large voltage)

The central idea is to combine these opposite temperature trends so that their changes approximately cancel.

For example, if a PTAT contribution increases by a certain amount as temperature rises, a suitably scaled CTAT contribution can decrease by approximately the same amount. The resulting sum changes much less.

Important: PTAT and CTAT do not automatically cancel. The circuit's resistor ratios, transistor characteristics, and operating currents must be designed to make the cancellation work.



How an NPN BJT is formed in CMOS ?

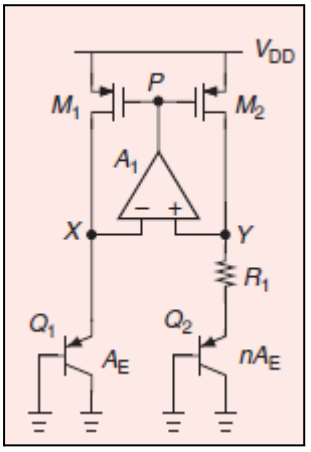
- Collector: An N-type region, potentially using a deep N-well structure in an appropriate device implementation.
- Base: A P-type region.
- Emitter: An N+ diffusion region.

Design Specifications

Parameters	Target Values
Technology	SKY130, 130 nm
Supply Voltage	1.8 V
Reference Voltage	~1.2 V
Temperature Range	-40 to 125 ℃
Power Consumption	<100 µW
Startup Time	<10 µs

1. Bandgap core topologies

1.1 Basic core



Main components

- Q_1 and Q_2: bipolar transistors with different emitter areas.
- M_1 and M_2: PMOS transistors that supply and mirror currents.
- R_1: converts the base-emitter voltage difference into a current.
- A_1: amplifier that regulates the core nodes.

How does it work ?

1) Q_1 has emitter area A_E, while Q_2 has emitter area nA_E. Their current densities differ when their currents are equal.
2) For matched current conditions, the magnitude of the difference is $\Delta V_{BE} = V_T \ln(n) \sim$ VPTAT
3) The amplifier adjusts the gate voltage of M_1 and M_2 so that $V_X \approx V_Y$
4) This feedback establishes the required relationship between the two BJT voltages and the voltage across R_1.
5) The PTAT current is $I_{PTAT} = \frac{V_T \ln(n)}{R_1}$
6) The base-emitter voltage V_{BE1} is present in the circuit and has a CTAT temperature trend.

Advantages

- Relatively simple starting circuit
- Generates a well-defined PTAT quantity
- Useful for establishing the BGR operating principle

Limitations

- Current increases with temperature
- Does not yet provide a temperature-independent output
- Requires an amplifier and suitable voltage headroom

Amplifier: An operational amplifier is often used in a conventional BGR to force appropriate operating conditions, such as equal voltages at two nodes, so the required PTAT current and voltage are generated accurately.

Voltage headroom: The circuit needs sufficient supply voltage for its transistors and other components to operate in their intended regions. If the supply is too low, the amplifier or transistors may not operate correctly.

Both BJT bases and emitters are grounded as drawn, so the base-emitter voltage difference is not directly a voltage between nodes X and Y. Instead, the different collector currents and branch connections establish the voltage across R_1.

Explanation with example

- Q_1: emitter area A_E
- Q_2: emitter area $8A_E$
- Both BJTs have the same collector current, $I_C = 10\mu A$.
- Temperature: $27^{\circ}C$.

For a BJT , $I_C = I_S e^{V_{BE}/V_T}$

A larger emitter area gives a larger saturation current I_S. Therefore, to conduct the same collector current, the larger BJT needs a lower V_{BE}.

VBE1 = 650 mV

VBE2 = 596.2 mV

$\Delta V_{BE} = 650 - 596.2 =$ **53.8 mV**

This agrees approximately with the theoretical equation: $\Delta V_{BE} = V_T \ln(8)$

If the circuit connections place the $53.8mV$ difference across R_1, and $R_1 = 10k\Omega$, then

$$I_{PTAT} = \frac{\Delta V_{BE}}{R_1}\ ;\ I_{PTAT} = \frac{53.8mV}{10k\Omega} = \mathbf{5.38}\ \mu A$$

PTAT Voltage Generation and Circuit Operation

1) The op-amp controls PMOS transistors M_1 and M_2 through feedback. Ideally, Vx ~ Vy
2) Look at the two terminals of R_1
    a) op terminal: connected to node Y.
    b) Bottom terminal: connected to the collector of Q_2; $V_{R_1} = V_Y - V_{C2}$. Vy = Vc1
    c) The BJT current-density difference gives $\Delta V_{BE} = V_T \ln(n)$
    d) In the diagram, the base and emitter terminals of both BJTs are drawn connected to ground. The collector terminals are connected to nodes X and the R_1 branch, respectively.
    e) For properly forward-biased BJTs, the base-emitter voltages depend on collector current and device area: $I_C \approx I_S e^{V_{BE}/V_T}$
    f) So ΔV_{BE} Appears across R1
    g) $I_{PTAT} = \frac{\Delta V_{BE}}{R_1} = \frac{V_T \ln(n)}{R_1}$

So, the intended PTAT operation may use this relationship, but it cannot be confirmed from the shown connections without verifying the original circuit

CTAT Generation

The base-emitter voltage of a BJT, V_{BE}, decreases as temperature increases.

For properly forward-biased BJTs, the base-emitter voltages depend on collector current and device area:

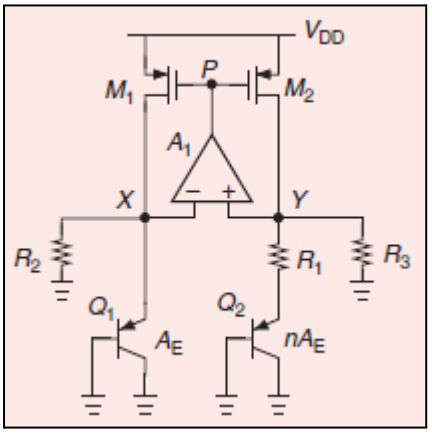
$I_C \approx I_S e^{V_{BE}/V_T}$

At node X = VBE1

At node Y = ΔV_{BE}

The voltage difference between the two BJT branches determines the voltage across R_1.

1.2 Core with R_2 and R_3



The next step is to modify the basic core so that its current becomes approximately temperature-independent.

Why are the additional resistors needed?

- In Basic Core, R_1 produces a PTAT current. To compensate for this, it is derived from the BJT base-emitter voltage, whose magnitude decreases with temperature.

How does it work ?

1) The PTAT voltage difference still appears across R_1.
2) The amplifier maintains $V_X \approx V_Y$.
3) The resistors connected to the ground introduce additional current contributions associated with the node voltages.
4) The PTAT and CTAT contributions are combined in the core current.
5) The resistor ratios are selected to make the net temperature coefficient close to zero.
6) CTAT is generated from the VBE of transistor with less emitter area of Q1 (have larger current) ICTAT was generated across R2 (we can also use Q2 for CTAT Generation)

$$|I_D| \approx \frac{V_T \ln(n)}{R_1} + \frac{|V_{BE2}|}{R_2}$$

R_2 provides an additional current path from node X to the ground.

Purpose: It introduces an additional current contribution at node X, affecting the current balance and the operating point of the circuit.

$$\frac{R_2}{R_1} V_T \ln(n) \approx 17 V_T \qquad \textit{assumes } R_2 = R_3.$$

Advantages

- Compensates for the temperature dependence of the PTAT current.

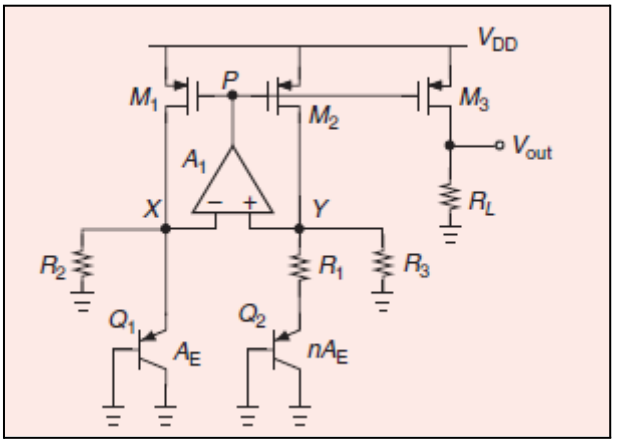
- Produces a much more temperature-stable current.
- Provides a suitable current for the output branch.

Limitations

- Compensation is approximate.
- Resistor mismatch and temperature coefficients affect the result.
- A flat current does not guarantee a flat output voltage if the current mirror introduces errors.

1.3. Core with output branch

- PMOS transistor M_3 and output resistor R_L added
- The compensated current is copied through M_3 into R_L, generating the output voltage (1.2 V).
- $V_{out} = \frac{R_L}{R_3}\left[\frac{R_3}{R_1}V_T \ln(n) + |V_{BE1}|\right]$



How does it work ?

1) The supply voltage is connected to the sources of PMOS transistors M_1, M_2, and M_3.
2) When the PMOS transistors turn on, conventional current flows from V_{DD} through their channels into their respective branches.
3) The currents through M_1 and M_2 pass through the resistor and BJT networks. Those networks determine the branch voltages and currents. (As explained in previous circuits)
4) The op-amp senses V_X and V_Y
    - If the voltages differ, its output changes.
    - This changes the common PMOS gate voltage V_P.
    - The PMOS currents change in response.
    - Negative feedback ideally settles at $V_X \approx V_Y$.
    - op-amp output is a control voltage
5) The gate of M_3 is connected to the same point P as the reference PMOS gates.
6) Therefore, M_3 receives the same gate voltage, not the current flowing through M_1 or M_2.
7) For matched PMOS transistors operating in saturation, $I_{M3} \approx I_{ref}$
8) If their W/L ratios are equal and their operating conditions are sufficiently similar.
9) $V_{out} = I_{M3}R_L$
10) $To\ get\ 1.2V$, RL = 1.2 / I ref (Iref is previously simulated)

Advantages

- Generates an actual voltage reference.
- Supports output levels below the conventional bandgap voltage.
- Provides the complete starting circuit for subsequent improvements.

Limitations

- Current-copy errors affect the output.
- Amplifier offset introduces output error.
- The output voltage can drift even when the core current is relatively stable

A. At $t = 0$: V_{DD} starts rising. The PMOS transistors and BJT branches begin establishing currents.

B. During startup: The op-amp compares V_X and V_Y and adjusts its output voltage, changing the common gate voltage V_P.

C. During settling: The branch currents and node voltages adjust in response to the feedback.

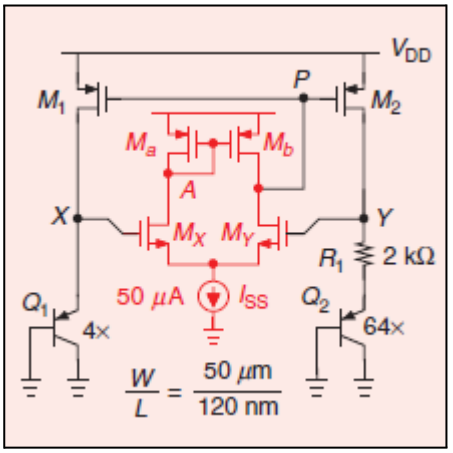
D. At steady state: Ideally, $V_X \approx V_Y$, the reference current is stable, and V_{out} approaches its final value.

2. The Amplifier topologies

For an ideal opamp

- V_X: voltage at node X.
- V_Y: voltage at node Y.
- $V_{diff} = V_Y - V_X$
- It amplifies this small difference and changes its output voltage V_P ; $V_P \approx A(V_Y - V_X)$
- PMOS gate voltage changes according to Vp

2.1 Five-transistor OTA



M_X and M_Y: NMOS differential pair

- These are the two NMOS transistors in the lower part of the red OTA circuit.
  - Gate of M_X connects to node X.
  - Gate of M_Y connects to node Y.
  - Their sources join at the common tail-current node.
  - Their drains connect to the PMOS active-load network.

They convert the voltage difference between X and Y into a difference in drain currents.

If $V_X = V_Y$, and the devices are matched and properly biased, the 50μA tail current is shared approximately equally: $I_{D,Mx} \sim I_{D,My} \sim 25\mu A$

This is the ideal matched operating point. When V_X and V_Y differ, the current split changes.

Important parameters for M_X and M_Y

- g_m: how strongly a small gate-voltage change alters drain current.
- V_{TH}: threshold voltage.
- V_{DS}: drain-source voltage.
- g_{ds}: output conductance, which influences voltage gain.

M_a and M_b: PMOS active-load/current-mirror pair

- These are the two PMOS transistors at the top of the red OTA circuit.
- They act as the active load for M_X and M_Y.
- Their jobs are:
    a. Provide the load for the NMOS differential pair.
    b. Convert the current imbalance into an output-voltage change.
    c. Help establish the OTA's voltage gain.
- The PMOS mirror makes the current in one branch influence the current supplied by the other branch. This allows the differential current to produce a useful single-ended output.
- The right-hand output branch connects to node P, which also controls the gates of the BGR's PMOS transistors M_1 and M_2.

The 50µA tail current source $I_{D,MX} + I_{D,MY} \approx I_{SS}$

The tail-current source affects:

- The transconductance of the input pair.
- The amplifier's gain and speed.
- Power consumption.
- The available voltage headroom.

The chain of action is:

1) M_X and M_Y sense V_X and V_Y.
2) Their drain currents change.
3) M_a and M_b convert the current imbalance into a change in P.
4) The voltage at P changes the gate-source voltages of M_1 and M_2.
5) Their branch currents change, correcting the voltage difference between X and Y.

If V_X increases relative to V_Y, more current flows through the corresponding input transistor.

If V_Y increases relative to V_X, the current split changes in the opposite direction.

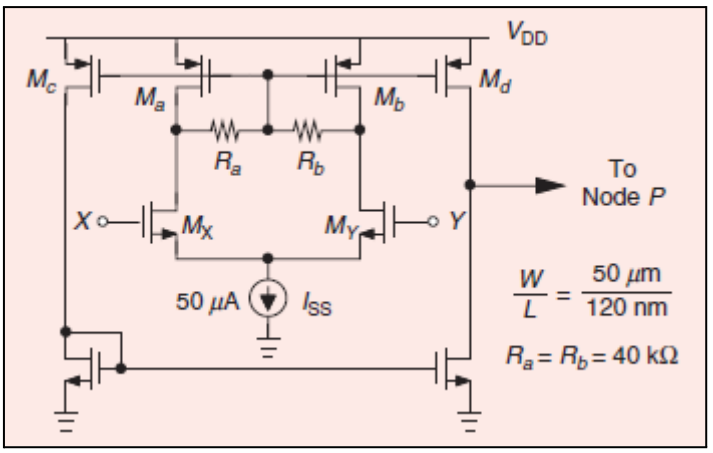
Advantages

- Simple circuit: Only five MOSFETs are required, making it relatively easy to design and integrate.
- Low power consumption: It uses a small bias current; the paper uses 50μA for this OTA.
- Compact layout: Fewer transistors generally mean less silicon area and fewer internal nodes.
- Differential input: It senses the voltage difference between X and Y, supporting negative feedback in the BGR.
- Active load: The PMOS load provides voltage gain without requiring large passive load resistors.
- Useful supply tracking: In this particular circuit, the PMOS active load helps node P track supply variations, contributing to supply rejection.

Limitations

- Limited voltage gain: The paper reports a gain of approximately 20, which is lower than the two-stage amplifier's gain.
- Limited PSRR: Supply disturbances can still affect the circuit, particularly at low temperatures and higher frequencies.
- Restricted input common-mode range: The NMOS input pair needs sufficient voltage headroom to operate correctly.
- Operating-region sensitivity: At low temperatures, the paper notes that the input transistors may approach the triode region, reducing amplifier gain.
- Mismatch and offset: Differences between transistor characteristics can create input-referred offset and affect BGR accuracy.
- Output swing limitations: The output voltage must remain within the range allowed by the PMOS load and the connected BGR transistors.

2.2 Two-stage CMOS op-amp



M_X and M_Y — NMOS differential input pair

- Gate of M_X connects to input X.
- Gate of M_Y connects to input Y.
- Their sources join at the common tail-current branch.
- Their drains connect to the upper PMOS network.
- Their function is to convert a small difference between the input voltages into a difference between drain currents.
- When $V_X = V_Y$, and the transistors are matched and properly biased, the tail current is divided approximately equally: $I_{D,MX} \approx I_{D,MY} \approx \frac{I_{SS}}{2}$
- If the input voltages become unequal, the current distribution changes. The loads above the pair convert this current change into a voltage signal.
- Parameters that matter: g_m, g_{ds}, threshold voltage, transistor mismatch and saturation headroom.

M_a and M_b — PMOS active loads

- Provide active loads for M_X and M_Y.
- Convert the differential pair's current changes into voltage changes.
- Contribute to the first stage's voltage gain.
- The first stage needs enough gain to turn a small input difference into a useful internal voltage change.
- A simplified gain estimate for a transistor gain stage is $|A_{v1}| \approx g_m R_{out}$
- where R_{out} is the effective small-signal resistance looking into the stage's output node. In the actual circuit, the transistor conductances and connections determine that resistance.

R_a and R_b — Bias-setting resistors

- These resistors are important because they help establish the output common-mode voltage of the first stage.
- The resistors help establish a suitable DC operating point so the transistors can work in the intended regions.
- They are not the main gain-producing components. Their primary role here is biasing.

M_c and M_d — PMOS devices in the second-stage and bias network

- They are part of the current-mirror and second-stage arrangement, rather than the central input pair.
- M_c participates in establishing a reference branch current.
- M_d provides the corresponding PMOS-side current for the output branch.
- Their operating points depend on the gate connections and the current established by the associated bias network.
- The current-mirror arrangement helps establish the current available to the second gain stage.
- Why is this needed? The second stage must have a properly established DC current before it can amplify the voltage signal from the first stage.

The two lower NMOS transistors - Current-mirror bias

- At the bottom of the schematic, there are two additional NMOS transistors.
- One is diode-connected: its gate and drain are connected together. The other shares its gate-bias connection, forming an NMOS current mirror.
- A diode-connected MOSFET develops a gate-source voltage corresponding to the current flowing through it. The mirror uses that voltage to establish a related current in the other branch.
- This helps set the second stage's bias current without requiring a separate ideal current source for every branch.
- The exact mirrored current depends on the transistor dimensions and operating conditions.

The overall gain is approximately the product of the individual stage gains, provided the stages are appropriately loaded.

Advantages

- Higher gain than the five-transistor OTA
- Better low-frequency PSRR in the BGR
- Bias currents established by transistor mirrors and resistors
- Can reduce the input error in the feedback loop

Limitations

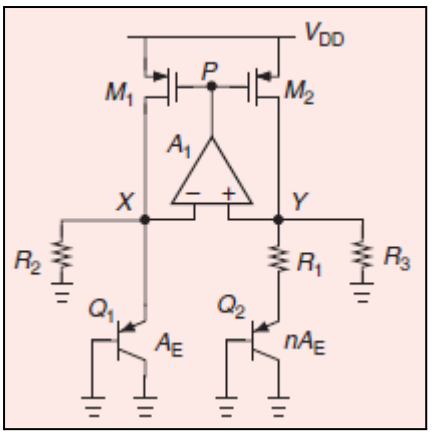
- More transistors and a more complex bias network
- Additional power consumption
- Sensitive to transistor mismatch and operating point
- More internal nodes can complicate frequency compensation and stability

Design Process

- M_X, M_Y: select g_m/I_D and current density for the required input-pair transconductance.
- M_a, M_b: evaluate g_m/g_{ds}, current-mirror matching and saturation headroom.
- M_c, M_d and lower NMOS mirror: choose dimensions for the required bias currents and output swing.
- R_a, R_b: verify the first-stage DC voltage and second-stage bias point.
- Complete amplifier: verify open-loop gain, output swing, power consumption, phase margin and PSRR across temperature.

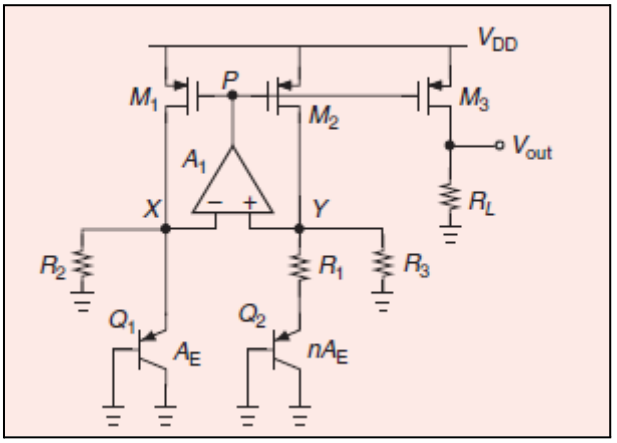
3. Output topologies

3.1 Current-output core



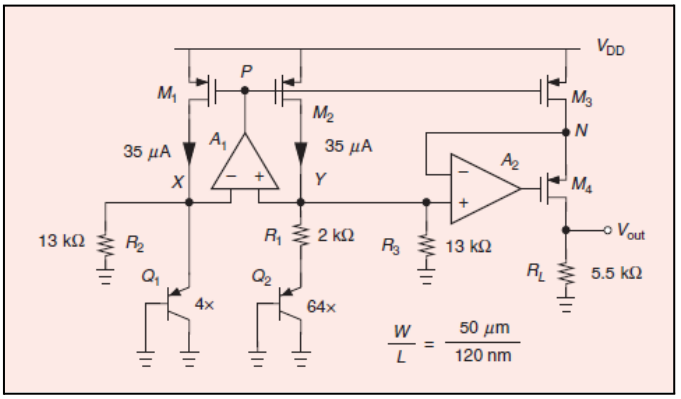
- This topology combines temperature-dependent effects so that the branch current becomes approximately independent of temperature.
- How it works :
    a. The BJT pair generates a voltage difference. The emitter areas of Q_1 and Q_2 differ by a factor n, producing a base-emitter voltage difference.
    b. Resistor R_1 generates a PTAT current. The voltage difference across R_1 produces a current that increases with absolute temperature.
    c. Resistors R_2 and R_3 introduce the CTAT contribution. The base-emitter voltage decreases as temperature increases, so its contribution opposes the PTAT effect.
    d. The currents combine. With the resistor values selected appropriately, the temperature dependence of the combined branch current is reduced.

3.2 Voltage-output core



- This topology converts the compensated current into a usable reference voltage by passing it through an output resistor.
- How it works :
    a. The BGR core generates the compensated current.
    b. PMOS transistor M_3 copies the current from the core branch into the output branch.
    c. The copied current flows through the load resistor R_L.
    d. The resistor converts the current into an output voltage.

3.3 Regulated-cascode voltage output



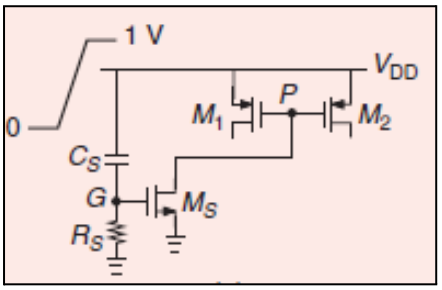
- The regulated-cascode output stage improves the accuracy of the copied current, reducing output-voltage variation.
- A1: Main BGR-core amplifier; regulates the core so that $V_X \approx V_Y$.
- M_2: Core current-setting transistor.
- M_3: Output current-copying transistor.
- M_4: Cascode transistor that helps control the output branch.
- A2: Additional amplifier that controls the gate of M_4.
- V_N: Internal node whose voltage is regulated to track V_Y.
- R_L: Converts the output current into V_{out}.

How A2 improves the output

1. A2 senses V_Y and V_N.
2. It compares these two voltages.
3. A2 adjusts the gate voltage of M_4.
4. This changes the operating point of M_4, driving V_N toward V_Y.
5. Keeping the relevant drain voltages similar reduces the current-copying error between M_2 and M_3.
6. Feedback condition $V_N \approx V_Y$

4. Startup topologies

4.1 Timing-based startup



Temporarily pull node P low during power-up so that the main BGR core starts operating.

Main components

- R_S and C_S: Create a time-dependent voltage at node G.
- M_S: Startup MOSFET that draws current from node P.
- Node P: Controls the main PMOS transistors in the BGR core.

How it works

1. Supply turns on. As V_{DD} rises, the voltage at node G changes through the timing network.
2. Startup transistor activates. When V_G reaches the required level, M_S turns on.
3. Node P is pulled low. M_S draws current from P, helping the main PMOS devices turn on and establish the core currents.
4. Startup action ends. After the supply stabilizes, the timing-node voltage returns toward its inactive state and M_S switches off.
5. Normal operation continues. The main BGR feedback loops maintain the reference.

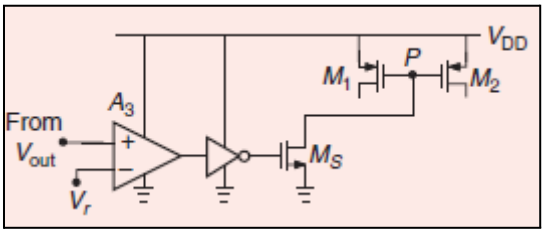
Advantages

- Conceptually simple.
- Does not require an output-voltage comparison amplifier.
- Can provide a startup pulse using an RC network.

Limitations

The timing must work across different supply-ramp rates, process variations and temperatures. The paper notes that a very slow supply ramp-for example, 1 ms-can require extremely large timing resistance and capacitance.

4.2 Voltage-comparison startup



Instead of estimating when startup should finish, this circuit checks whether the BGR output has actually risen.

Main components

- A3: Startup OTA/comparator.
- V_{out}: Actual BGR output voltage.
- V_r: Startup comparison reference.
- M_S: Startup transistor that draws current from node P.

How it works

- At power-up, the output may be zero. The main BGR core might not yet be operating.
- A3 compares V_{out} with V_r. The comparison indicates whether the output is sufficiently low.
- If the output is too low, A3 activates M_S. The startup transistor draws current from P, helping the main core turn on.
- The output rises. As the BGR establishes its normal operating state, V_{out} approaches its intended value.
- Startup assistance is removed. Once the comparison no longer indicates an insufficient output, A3 disables the startup action.

Advantages

- Responds to the actual output condition rather than relying solely on an RC delay.
- Can accommodate variation in startup time.
- Avoids the need for a very large timing network designed for a slow supply ramp.

Limitations

- Requires an additional amplifier and startup transistor.
- The comparison reference and amplifier offset affect the startup threshold.
- The startup circuit must disengage properly once the BGR is operating.

SUITABLE COMBINATION OF TOPOLOGIES FOR THIS SPECIFICATIONS

Core Topology : Core with R_2 and R_3

The BJT pair generates PTAT and CTAT contributions. The resistor network combines them to make the core current less temperature-dependent.

Main Amplifier Topology : Five-transistor OTA

The five-transistor OTA is compact and comparatively simple to bias. It senses V_X and V_Y and adjusts the PMOS gate voltage through feedback.

Output Topology : Core with output branch 1.3

Startup Topology : Voltage-comparison startup

A3 detects when the output is too low and activates the startup transistor. It responds to the circuit's actual output condition rather than relying only on a timing delay.

Reference: The Design of a Low-Voltage Bandgap Reference, Behzad Razavi